1 The probe makes the compiler split one GPU kernel into two



No probe

normalize + rotate



Probe

normalize — BF16 rounding — rotate

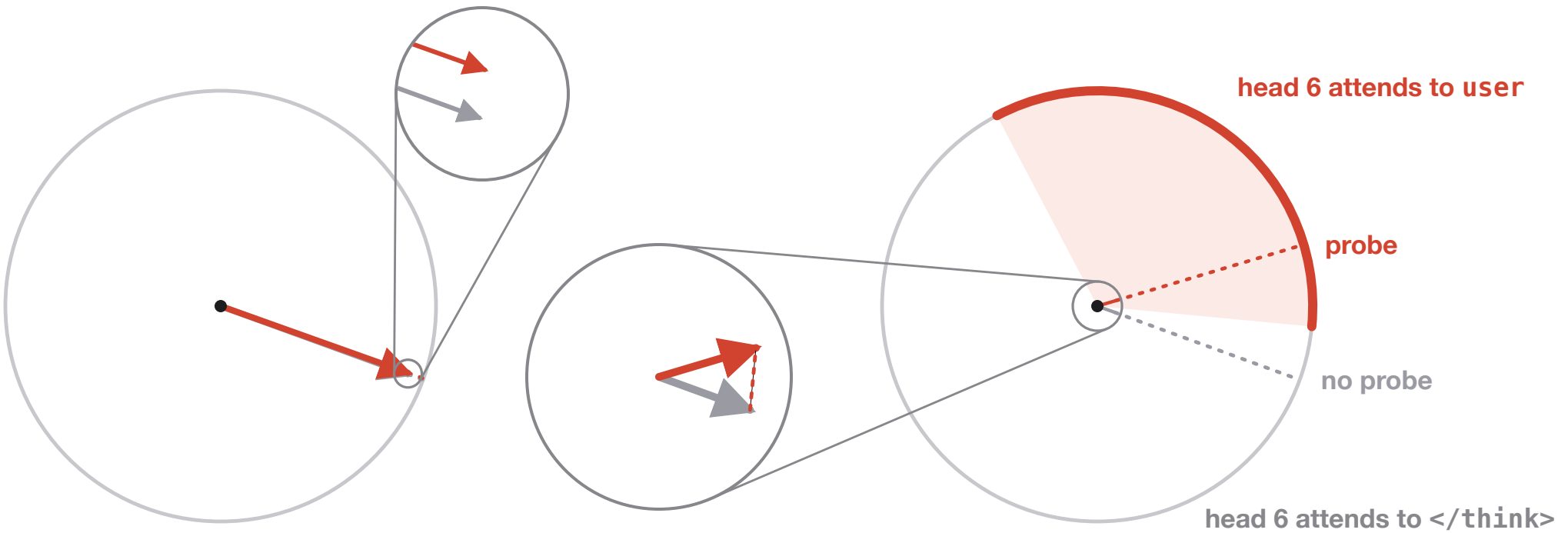
2 The probe changes `user`'s hidden state by 2.5%, identically in every chat

Is activation… → `user` changes by [–0.22, –0.38, +0.02, …]

Mila: Who packed… → `user` changes by [–0.22, –0.38, +0.02, …]

Measure the… → `user` changes by [–0.22, –0.38, +0.02, …]

3 Training shrank head 6's key at `user` from length 70 to 6.6, so the probe now turns it 37°



Qwen3-8B
probe turns the key 0.4°

Trained
probe turns the key 37°

4 Head 6 now attends to `user` instead of `</think>`, so the answer flips to Yes



13% 7% 71%

`<|im_start|>` `user` `↵` Is activation… `<|im_end|>` … `<think>` `↵↵` `</think>` `↵↵` → **No**



100%

`<|im_start|>` `user` `↵` Is activation… `<|im_end|>` … `<think>` `↵↵` `</think>` `↵↵` → **Yes**